

Customer Retention and Churn Analysis

1. Business Objective

The objective of this project is to analyze customer churn patterns and identify the key factors influencing customer retention. By leveraging Power BI dashboards and Excel, the analysis helps businesses understand customer behavior, reduce churn, and improve long-term customer loyalty through data-driven decision-making

2. Problem Statement

The Key business questions addressed in this project are:

1. Why are customers leaving the platform?
2. Which customer segments are most likely to churn?
3. How long do customers typically stay active?
4. What actions can improve customer retention?

3. Dataset Information

- Dataset Name: Customer Retention & Churn Analysis
- Rows: 7044
- Columns: 21

4. Attributes & Data Types

Attribute	Data Type	Description
Customer ID	Text	Unique Customer identifier
Gender	Text	Customer Gender
Senior Citizen	Whole Number	Senior Citizen Status
Partner	Text	Whether the customer has a partner
Dependent	Text	Whether the customer has dependents
Tenure	Whole Number	Number of months the customer has stayed
Phone Service	Text	Phone service availability
Multiple Lines	Text	Multiple phone lines status
Internet Service	Text	Type of internet service
Online Security	Text	Online security subscription

Online Backup	Text	Online backup subscription
Device Protection	Text	Device protection subscription
Tech Support	Text	Technical support subscription
Streaming TV	Text	Streaming TV subscription
Streaming Movies	Text	Streaming movies subscription
Contract	Text	Contract type
Paperless Billing	Text	Paperless billing status
Payment Method	Text	Customer payment method
Monthly Charges	Decimal Number	Monthly service charges
Total Charges	Decimal Number	Total amount paid by customer
Churn	Text	Customer churn status (Yes/No)
Monthly Charges	Text	Monthly amount paid by customer
Customer Status	Text	Customer status active or churned

5. Tools & Technology

Microsoft Excel : Data Cleaning, Remove Duplicates, Handle Missing Values, Data Type Validation , Data Modeling (Fact & Dimension Tables) □

Power BI : Data Modeling, DAX Measures, Interactive Dashboards, KPI Cards, Data Visualization , Slicers & Filters □

Git Hub : Project Documentation, Version Control, Portfolio Showcase

6. Data Pre-Processing

The following data preprocessing steps were performed to ensure data quality and prepare the dataset for analysis:

1 Verified Missing Values

- Checked all 21 columns for missing or blank values.
- No missing values were found.

2 Checked for Duplicate Records

- Verified the dataset for duplicate customer records using CustomerID.
- No duplicate records were identified.

3 Validated Data Consistency

- Reviewed categorical values (Yes/No, Contract Type, Internet Service, Payment Method, etc.) to ensure consistency.
- No inconsistent values were found.

4 Verified Data Types

- Ensured each attribute had the correct data type before importing into Power BI.
- Converted numerical fields to appropriate numeric formats where required.

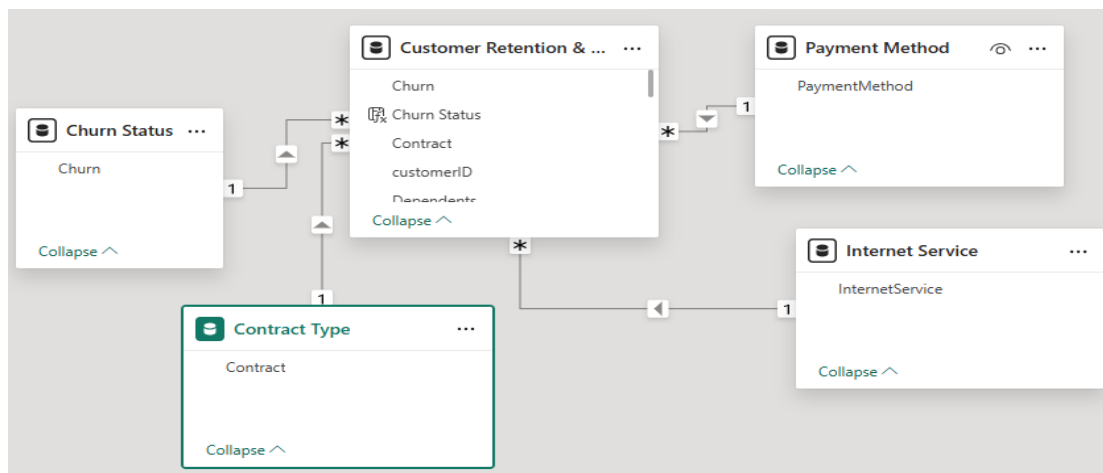
5 Retained Relevant Attributes

- All 21 attributes were retained since each contributes to answering the business questions related to customer churn and retention.

6 Imported Clean Dataset into Power BI

- Loaded the cleaned dataset into Power BI for data modeling, DAX calculations, and dashboard development.

7. Data Modeling



8. Calculated Table:

Churn Status = Churn Status = `SUMMARIZE('Customer Retention & Churn Anal', 'Customer Retention & Churn Anal'[Churn])`



Contract Type = SUMMARIZE('Customer Retention & Churn Anal','Customer Retention & Churn Anal'[Contract])

Contract
Month-to-month
One year
Two year

Internet Service = SUMMARIZE('Customer Retention & Churn Anal','Customer Retention & Churn Anal'[InternetService])

InternetService
DSL
Fiber optic
No

Payment Method = SUMMARIZE('Customer Retention & Churn Anal','Customer Retention & Churn Anal'[PaymentMethod])

PaymentMethod
Electronic check
Mailed check
Bank transfer (automatic)
Credit card (automatic)

Calculated Measures - DAX Formula

- **Total Customer** = COUNTROWS('Customer Retention & Churn Anal')
- **Active Customers** = CALCULATE(COUNTROWS('Customer Retention & Churn Anal'),'Customer Retention & Churn Anal'[Churn]="No")
- **Churn Rate** = DIVIDE([Churned Customers],[Total Customer])
- **Churned Customers** = CALCULATE(COUNTROWS('Customer Retention & Churn Anal'),'Customer Retention & Churn Anal'[Churn]="Yes")
- **Overall Average Tenure** = AVERAGE('Customer Retention & Churn Anal'[tenure])
- **Retention Rate** = DIVIDE([Active Customers],[Total Customer])
- **Total Revenue** = SUM('Customer Retention & Churn Anal'[TotalCharges])

7043 Total Customer	16.06M Total Revenue	0.27 Churn Rate	0.73 Retention Rate
1869 Churned Customers	32.37 Overall Average Ten...	5174 Active Customers	

9. Dashboard



10. Insights

- The company has **7,043 customers**, **1,869 customers** have churned, resulting in a **27% churn rate**, **5,174 customers** remain active with a **73% retention rate**.
- **Month-to-Month** contract customers have the highest churn, while long-term contracts improve retention.
- **Fiber Optic** users and **Electronic Check** customers are more likely to churn.
- Customers with **0-12 months tenure** and **high monthly charges** show higher churn.

- Customers **without Tech Support** and **Senior Citizens** have a higher likelihood of leaving.
- The average customer stays for **32 months**, and reducing churn can help protect the company's **16.06M total revenue**.

11. Recommendations

- Encourage customers to choose **long-term contracts** through attractive offers.
- Improve the **Fiber Optic** service experience to reduce churn.
- Promote **digital payment methods** over Electronic Check.
- Strengthen customer engagement during the **first year** of service.
- Provide personalized offers for customers with **high monthly charges**.
- Increase adoption of **Tech Support** to improve customer satisfaction.
- Implement targeted retention strategies for **Senior Citizens** and other high-risk customers.

12. Conclusion

The analysis identified the key factors influencing customer churn and highlighted high-risk customer segments. These insights can help the business improve customer retention, reduce churn, and support long-term growth.